MILITARY SPECIFICATION

SWITCH, COMBINED OPTION SELECTOR: M167E1

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switch entitled Combined Option Selector Switch: M167E1 which is used in the Lance warhead section to select the mode of explosion.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-6021	- Casting, Classification and Inspection.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification, Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-456	- Electronic Parts, Date and Source Coding for.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1135

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DRAWINGS

ORDNANCE CORPS

8880138	- Housing, Switch.
9316971	- Switch.
8880197	- Switch.
9316975	- Switch, Combined Option Selector: M167E1
9236827	- Housing, Casting.
9236832	- Actuator Splicing Subassembly.
ET-9316975	- Equipment Tabulation.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switch shall comply with all requirements specified on the drawing (dwg) 9316975 with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-N-A. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

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3.5 Circuit resistance. The circuit resistance between the terminal pins listed in Tables I and II shall not exceed 0.050 ohms when tested in accordance with 4.4.1. Record actual values in accordance with 3.13.

3.6 Insulation resistance. The insulation resistance between each pin and all other pins, listed below at the designated switch positions, and the case shall not be less then 50 megohms when subjected to a potential of 500 volts direct current (vdc) with the current limited to five (5) milliamperes (ma) when tested as specified in 4.4.2. Record actual values in accordance to 3.13.

a. No. 1 Switch Positions (J1 Connector)

(1) position S:

Pin to Pin

1 to 2 thru 31, 33, 34, 36, 37, case
2 to 3 thru 31, 33, 34, 36, 37, case
3 to 4 thru 31, 33, 34, 36, 37, case
4 to 5 thru 31, 33, 34, 36, 37, case
5 to 6 thru 31, 33, 34, 36, 37, case
6 to 7 thru 31, 33, 34, 36, 37, case
7 to 8 thru 31, 33, 34, 36, 37, case
8 to 9 thru 31, 33, 34, 36, 37, case
9 to 10 thru 31, 33, 34, 36, 37, case
10 to 11 thru 31, 33, 34, 36, 37, case
11 to 12 thru 31, 33, 34, 36, 37, case
12 to 13 thru 31, 33, 34, 36, 37, case
13 to 14 thru 31, 33, 34, 36, 37, case
14 to 15 thru 31, 33, 34, 36, 37, case
15 to 16 thru 31, 33, 34, 36, 37, case
16 to 17 thru 31, 33, 34, 36, 37, case
17 to 18 thru 31, 33, 34, 36, 37, case
18 to 19 thru 31, 33, 34, 36, 37, case
19 to 20 thru 31, 33, 34, 36, 37, case
20 to 21 thru 31, 33, 34, 36, 37, case
21 to 22 thru 31, 33, 34, 36, 37, case
22 to 23 thru 31, 33, 34, 36, 37, case
23 to 24 thru 31, 33, 34, 36, 37, case
24 to 25 thru 31, 33, 34, 36, 37, case
25 to 26 thru 31, 33, 34, 36, 37, case
26 to 27 thru 31, 33, 34, 36, 37, case
27 to 28 thru 31, 33, 34, 36, 37, case
28 to 29 thru 31, 33, 34, 36, 37, case
29 to 30, 31, 33, 34, 36, 37, case
30 to 31, 33, 34, 36, 37, case
31 to 33, 34, 36, 37, case
33 to 34, 36, 37, case
34 to 36, 37, case
36 to 37, case
37 to case

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All unused pins shall be connected together and tested as one pin to case.

(2) position LG:

Pin to Pin

1 to 4, 5, 9, 10, 12, 13, 14, 16, 18, 19, 20, 25 thru 31, 36, case
4 to 5, 9, 10, 12, 13, 14, 16, 18, 19, 20, 25 thru 31, 36, case
5 to 9, 10, 12, 13, 14, 16, 18, 19, 20, 25 thru 31, 36, case
9 to 10, 12, 13, 14, 16, 18, 19, 20, 25 thru 31, 36, case
10 to 12, 13, 14, 16, 18, 19, 20, 25 thru 31, 36, case
12 to 13, 14, 16, 18, 19, 20, 25 thru 31, 36, case
13 to 14, 16, 18, 19, 20, 25 thru 31, 36, case
14 to 16, 18, 19, 20, 25 thru 31, 36, case
16 to 18, 19, 20, 25 thru 31, 36, case
18 to 19, 20, 25 thru 31, 36, case
19 to 20, 25 thru 31, 36, case
20 to 25 thru 31, 36, case
25 to 26 thru 31, 36, case
26 to 27 thru 31, 36, case
27 to 28 thru 31, 36, case
28 to 29 thru 31, 36, case
29 to 30, 31, 36, case
30 to 31, 36, case
31 to 36, case
36 to case

All unused pins shall be connected together and tested as one pin to case.

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Table I

No. 1 Switch positions (J1 connector)									
HG P/P	Test Code No.	LG P/P	Test Code No.	G P/P	Test Code No.	H P/P	Test Code No.	L P/P	Test Code No.
16-23	004001	16-23	004015	16-23	004029	16-23	004035	16-23	004047
15-4	004002	15-4	004016	17-5	004030	15-4	004036	15-4	004048
30-29	004003	30-6	004017	21-9	004031	30-29	004037	30-6	004049
14-11	004004	14-11	004018	2-25	004032	14-11	004038	14-11	004050
17-5	004005	17-5	004019	22-10	004033	17-5	004039	17-5	004051
18-13	004006	18-7	004020	3-26	004034	18-13	004040	18-7	004052
21-9	004007	21-9	004021			21-9	004041	21-9	004053
2-25	004008	2-25	004022			19-28	004042	19-33	004054
19-28	004009	19-33	004023			31-12	004043	31-8	004055
22-10	004010	22-10	004024			1-36	004044	1-24	004056
31-12	004011	31-8	004025			3-26	004045	3-26	004057
1-36	004012	1-24	004026			20-27	004046	20-34	004058
3-26	004013	3-26	004027						
20-27	004014	20-34	004028						

Table II

No. 2 Switch positions (J2 connector)							
A P/P	Test Code No.	B P/P	Test Code No.	C P/P	Test Code No.	X P/P	Test Code No.
Y-E	004059	Y-T	004065	Y-G	004069	Y-A	004073
J-V	004060	J-U	004066	J-H	004070	J-B	004074
L-P	004061	L-C	004067	L-F	004071	L-M	004075
K-N	004062	K-D	004068	K-W	004072	K-R	004076
a-b*	004063						
J2-c to J1-37	004064						

*Resistance shall be 1.25 ± 0.20 ohms

P/P means pin-to-pin

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(3) position HG

Pin to Pin

2 to 3, 5, 6, 7, 8, 14, 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
3 to 5, 6, 7, 8, 14, 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
5 to 6, 7, 8, 14, 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
6 to 7, 8, 14, 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
7 to 8, 14, 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
8 to 14, 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
14 to 15, 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
15 to 16, 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
16 to 18, 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
18 to 19, 20, 21, 22, 24, 30, 31, 33, 34, 36, case
19 to 20, 21, 22, 24, 30, 31, 33, 34, 36, case
20 to 21, 22, 24, 30, 31, 33, 34, 36, case
21 to 22, 24, 30, 31, 33, 34, 36, case
22 to 24, 30, 31, 33, 34, 36, case
24 to 30, 31, 33, 34, 36, case
30 to 31, 33, 34, 36, case
31 to 33, 34, 36, case
33 to 34, 36, case
34 to 36, case
36 to case

All unused pins shall be connected together and tested as one pin to case.

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(4) position G

Pin to Pin

1 to 4 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
4 to 5 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
5 to 6 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
6 to 7 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
7 to 8 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
8 to 9 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
9 to 10 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
10 to 11 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
11 to 12 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
12 to 13 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
13 to 14 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
14 to 15 thru 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
15 to 16, 18, 19, 20, 24 thru 31, 33, 34, 36, case
16 to 18, 19, 20, 24 thru 31, 33, 34, 36, case
18 to 19, 20, 24 thru 31, 33, 34, 36, case
19 to 20, 24 thru 31, 33, 34, 36, case
20 to 24 thru 31, 33, 34, 36, case
24 to 25 thru 31, 33, 34, 36, case
25 to 26 thru 31, 33, 34, 36, case
26 to 27 thru 31, 33, 34, 36, case
27 to 28 thru 31, 33, 34, 36, case
28 to 29 thru 31, 33, 34, 36, case
29 to 30, 31, 33, 34, 36, case
30 to 31, 33, 34, 36, case
31 to 33, 34, 36, case
32 to 33, 34, 36, case
33 to 34, 36, case
34 to 36, case
36 to case

All unused pins shall be connected together and tested as one pin to case.

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(5) position L

Pin to Pin

2 to 3, 4, 6, 7, 8, 9, 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
3 to 4, 6, 7, 8, 9, 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
4 to 6, 7, 8, 9, 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
6 to 7, 8, 9, 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
7 to 8, 9, 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
8 to 9, 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
9 to 10, 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
10 to 12, 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
12 to 13, 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
13 to 14, 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
14 to 17, 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
17 to 22, 23, 24, 25, 27, 28, 29, 33, 34, 36, case
22 to 23, 24, 25, 27, 28, 29, 33, 34, 36, case
23 to 24, 25, 27, 28, 29, 33, 34, 36, case
24 to 25, 27, 28, 29, 33, 34, 36, case
25 to 27, 28, 29, 33, 34, 36, case
27 to 28, 29, 33, 34, 36, case
28 to 29, 33, 34, 36, case
29 to 33, 34, 36, case
33 to 34, 36, case
34 to 36, case
36 to case

All unused pins shall be connected together and tested as one pin to case.

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(6) position H

Pin to Pin

1 to 2, 6, 7, 8, 10, 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
2 to 6, 7, 8, 10, 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
6 to 7, 8, 10, 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
7 to 8, 10, 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
8 to 10, 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
10 to 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
11 to 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
15 to 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
16 to 17, 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
17 to 18, 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
18 to 19, 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
19 to 20, 21, 22, 24, 25, 26, 30, 31, 33, 34, case
20 to 21, 22, 24, 25, 26, 30, 31, 33, 34, case
21 to 22, 24, 25, 26, 30, 31, 33, 34, case
22 to 24, 25, 26, 30, 31, 33, 34, case
24 to 25, 26, 30, 31, 33, 34, case
25 to 26, 30, 31, 33, 34, case
26 to 30, 31, 33, 34, case
30 to 31, 33, 34, case
31 to 33, 34, case
33 to 34, case
34 to case

All unused pins shall be connected together and tested as one pin to case.

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b. No. 2 Switch Positions (J2 connector)

(1) position A

Pin to Pin

A to B, C, D, E, F, G, H, J, K, L, M, R, T, U, W, a, c, case
B to C, D, E, F, G, H, J, K, L, M, R, T, U, W, a, c, case
C to D, E, F, G, H, J, K, L, M, R, T, U, W, a, c, case
D to E, F, G, H, J, K, L, M, R, T, U, W, a, c, case
E to F, G, H, J, K, L, M, R, T, U, W, a, c, case
F to G, H, J, K, L, M, R, T, U, W, a, c, case
G to H, J, K, L, M, R, T, U, W, a, c, case
H to J, K, L, M, R, T, U, W, a, c, case
J to K, L, M, R, T, U, W, a, c, case
K to L, M, R, T, U, W, a, c, case
L to M, R, T, U, W, a, c, case
M to R, T, U, W, a, c, case
R to T, U, W, a, c, case
T to U, W, a, c, case
U to W, a, c, case
W to a, c, case
a to c, case
c to case

All unused pins shall be connected together and tested as one pin to case.

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(2) position B

Pin to Pin

A to B, C, D, E, F, G, H, J, M, N, P, R, T, V, W, case
B to C, D, E, F, G, H, J, M, N, P, R, T, V, W, case
C to D, E, F, G, H, J, M, N, P, R, T, V, W, case
D to E, F, G, H, J, M, N, P, R, T, V, W, case
E to F, G, H, J, M, N, P, R, T, V, W, case
F to G, H, J, M, N, P, R, T, V, W, case
G to H, J, M, N, P, R, T, V, W, case
H to J, M, N, P, R, T, V, W, case
J to M, N, P, R, T, V, W, case
M to N, P, R, T, V, W, case
N to P, R, T, V, W, case
P to R, T, V, W, case
R to T, V, W, case
T to V, W, case
V to W, case
W to case

All unused pins shall be connected together and tested as one pin to case.

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(3) position C

Pin to Pin

A to B, C, D, E, F, G, H, K, M, N, P, R, T, U, V, case
B to C, D, E, F, G, H, K, M, N, P, R, T, U, V, case
C to D, E, F, G, H, K, M, N, P, R, T, U, V, case
D to E, F, G, H, K, M, N, P, R, T, U, V, case
E to F, G, H, K, M, N, P, R, T, U, V, case
F to G, H, K, M, N, P, R, T, U, V, case
G to H, K, M, N, P, R, T, U, V, case
H to K, M, N, P, R, T, U, V, case
K to M, N, P, R, T, U, V, case
M to N, P, R, T, U, V, case
N to P, R, T, U, V, case
P to R, T, U, V, case
R to T, U, V, case
T to U, V, case
U to V, case
V to case

All unused pins shall be connected together and tested as one pin to case.

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(4) position X

Pin to Pin

A to B, C, D, E, F, G, H, K, L, N, P, T, U, V, W, case
B to C, D, E, F, G, H, K, L, N, P, T, U, V, W, case
C to D, E, F, G, H, K, L, N, P, T, U, V, W, case
D to E, F, G, H, K, L, N, P, T, U, V, W, case
E to F, G, H, K, L, N, P, T, U, V, W, case
F to G, H, K, L, N, P, T, U, V, W, case
G to H, K, L, N, P, T, U, V, W, case
H to K, L, N, P, T, U, V, W, case
K to L, N, P, T, U, V, W, case
L to N, P, T, U, V, W, case
N to P, T, U, V, W, case
P to T, U, V, W, case
T to U, V, W, case
U to V, W, case
V to W, case
W to case

All unused pins shall be connected together and tested as one pin to case.

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3.7 Indexing. Indexing shall be accomplished by means of a positive detent mechanism. The detent mechanism shall require an input torque of eight plus two minus three inch pounds when selector is rotated through each position both clockwise and counter clockwise. Testing shall be in accordance with 4.4.3. Record torque in accordance with 3.13.

No. 1 Switch			No. 2 Switch		
Position	Rotation	Test Code No.	Position	Rotation	Test Code No.
S to HG	CW	006001	A to B	CW	006013
HG to LG	CW	006002	B to C	CW	006014
LG to G	CW	006003	C to X	CW	006015
G to H	CW	006004	X to C	CCW	006016
H to L	CW	006005	C to B	CCW	006017
L to S	CW	006006	B to A	CCW	006018
S to L	CCW	006007			
L to H	CCW	006008			
H to G	CCW	006009			
C to LG	CCW	006010			
LG to HG	CCW	006011			
HG to S	CCW	006012			

3.8 Switch Assembly leakage. The unit shall be purged and filled with 90% nitrogen and 10% helium. The leak rate shall be 1 X 10^{-4} CC/sec maximum at an external pressure of 10 microns when tested as specified in 4.4.4. Record the leak rate in accordance with 3.13. The code no. is 007003.

3.9 Casting. The contractor shall furnish objective evidence (see 6.2) that the casting complies with dwg. 9236827. Testing shall be as specified in 4.4.5.

3.10 Environmental testing. The switch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The switch shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.6. Tests shall be conducted in the order specified.

3.10.1 Thermal shock. The switch shall withstand thermal shocks between +160 ± 5°F to -65 ± 5°F for three complete cycles without evidence of external damage (see 6.5) and shall comply with the requirements of 3.5, 3.6 and 3.7 after testing as specified in 4.4.6.1.

3.10.2 Mechanical shock. The switch shall withstand exposure to shock of 50 ± 5 gravity units (g's) peak amplitude, half sine durations of 11 ± 1 milliseconds, applied once in each direction along each of the three (3) perpendicular axes (total of 6 shocks) without evidence of external damage (see 6.5) and shall comply with the requirements of 3.5, 3.6 and 3.7 after testing as specified in 4.4.6.2.

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3.10.3 Transportation vibration. The switch shall withstand vibration tests along each of three (3) mutually perpendicular axes as specified in Table III without evidence of external damage (see 6.5) and shall be tested as specified in 4.4.6.3. Frequency shall be cycled at a logarithmic rate between 5-500-5 in fifteen (15) minutes. At the completion of this test, the switch shall comply with 3.5, 3.6 and 3.7.

Table III

Vibration test schedule			
Frequency (Hz)	Amplitude	Sweep Cycles	
		Ambient -65°F	+160 + 5°F
5-15	0.4" da	1	1
15-80	5 g	1	1
80-115	0.015" da	1	1
115-500	10 g	1	1

3.10.4 Operational vibration. The switch shall be vibrated at room temperature, -40 + 5°F, and +160 + 5°F in each of three mutually perpendicular axes as specified below without evidence of external damage (see 6.5). During vibration exposure, contact chatter shall be monitored and shall not exceed 10 microseconds with a contact load of 100 + 10 milliamperes. After testing, the switch shall comply with 3.11, 3.5, 3.6 and 3.7. The vibration test shall be conducted as specified in 4.4.6.4.

a. .06 g /Hertz (Hz) power spectral density (9.3 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 6 decibels per octave so that the total frequency enveloped is 50 to 2000 Hz for 10 seconds. The tolerance level shall be + 3 decibels (Db) between 100 to 1000 Hz and + 6 Db between 1000 to 2000 Hz.

b. .02 g /Hz power spectral density (5.3 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 6 Db so that the frequency enveloped is 50 to 2000 Hz for two (2) minutes. The tolerance level shall be + 3 Db between 100 to 2000 Hz.

3.11 Current carrying capacity. The maximum allowable voltage drop between terminals listed in Table IV shall not exceed 400 millivolts. The unit shall also be capable of carrying a resistance load of 20 + 2 amperes for 200 + 5 milliseconds. After being tested as specified in 4.4.7, the unit shall be tested for compliance with 3.5 and 3.6. Record voltage drop in accordance with 3.13.

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Table IV

Connector	Switch	Position	Terminals	Test Code No.
J1	1	HG	11-14	013001
J2	2	B	A-M	013002

3.12 Life cycling. The unit shall be able to withstand 300 cycles of each switch without degradation of 3.11, 3.5, 3.6 and 3.7. Testing shall be in accordance with 4.4.8.

3.13 Test codes. The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.14 Marking. Switches shall be marked in accordance with MIL-STD-130 with the following information:

a. Manufacturers name or trademark.
b. Government drawing number, type designation, and serial number (see 3.4).
c. Date code in accordance with MIL-STD-456.
d. Switch terminals shall be marked as specified in the applicable detail specification or drawing.

The finish or any dimension of the item shall not be altered or damaged by application of marking information. The material used for marking shall not be subject to corrosion, fungus, flaking or similar adverse action which could prevent proper functioning of the switches.

3.15 Soldering. All soldering shall be in compliance with MIL-STD-1460(MU).

3.16 Workmanship.

3.16.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.16.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.16.3 Burrs. Parts shall be free from burrs.

3.16.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.16.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.16.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (see dwg 9316975) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table V. Tests shall be conducted in order specified in Table V.

Table V

First article inspection and tests			
Inspection or Test	Testing Sequence Unit Nos.	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	all		X
2. Thermal shock (see 3.10.1)	1,4,6,7	X	
3. Transportation vibration (see 3.10.3)	2,3,5,8		X
4. Mechanical shock (see 3.10.2)	1,4,5,8	X	
5. Operational vibration @ +160°F (see 3.10.4)	1,3,6,8	X	
6. Operational vibration @ -40°F (see 3.10.4)	1,2,7,8	X	
7. Circuit resistance (see 3.5)	all		X
8. Insulation resistance (see 3.6)	all		X
9. Indexing (see 3.7)	all		X
10. Combined option selector switch leakage (see 3.8)	all		X
11. Current carrying capacity (see 3.11)	all		X
12. Life cycling (see 3.12)	1,3,4,7	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer of raw material or components, one production method or manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VI andTable VII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VI
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273

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Table VI (Cont'd)
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Table VII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch (see dwg 9316971 and dwg 8880197).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Soldering in accordance with MIL-STD-1460(MU), incorrect	Visual
Minor:	None defined.	

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4.3.2.1.2 Housing, switch (see dwg 8880138).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Flatness of mounting surface incorrect	Gage
102.	Location of mounting holes, incorrect	Gage
103.	Diameter of mounting holes, incorrect	Gage
104.	'O' ring area surface finish (6 places), incorrect	Visual
105.	Objective evidence of radiograph inspection (see 9236827)	Visual
Minor:	None defined.	

4.3.2.1.3 Switch combined option selector: M167E1 (see dwg 9316975).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Sealant on 40 screws missing	Visual
102.	Torque on 40 screws incorrect	Gage
103.	Torque on 5 nuts incorrect	Gage
104.	Major keyway parallel to datum surface (2)	Gage
105.	Any components missing, damaged or improperly assembled	Visual
106.	Cable connector interface, incorrect	Gage
107.	Objective evidence of performance acceptance (dwg 9236832)	Visual
108.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Name plate missing, illegible or incorrect	Visual
202.	Evidence of poor workmanship (see 3.16)	Visual

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4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.16)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major
c. Indexing	Major
d. Combined option selector switch leakage	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance, insulation resistance, indexing, and switch assembly leakage. Each switch shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.5 through 3.8, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-9316975. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The switch shall be tested on an approved four terminal volt-ammeter measuring equipment to determine compliance with 3.5. The magnitude of the direct current (dc) through the contacts shall be 100 ± 10 milliamperes (ma) and the open circuit test voltage shall not exceed 32 volts direct current (vdc). Separate voltage and current hookup wiring shall be used at each terminal in order not to include the effect of hookup wiring on measurement values. Activation of switches for purpose of improving contact surfaces prior to actual measurement of resistance shall not be permitted.

4.4.2 Insulation resistance. The insulation resistance shall be measured on approved equipment at a potential of 500 ± 50 vdc and current shall not exceed 5 milliamperes (ma). The potential shall be applied for a minimum of 5 seconds while testing to determine compliance with 3.6.

4.4.3 Indexing. The switch shall be mounted on a suitable fixture and be engaging an approved wrench, the torque required to rotate the switch from one position to another throughout the switch range and in either direction shall comply with the requirements of 3.7.

4.4.4 Switch assembly leakage. The switch shall be tested in accordance with MIL-STD-202, Method 112, Test Condition C, Procedure IV with a reduced pressure of 10 micron for one minute minimum; perforation of the unit is not a requirement.

4.4.5 Casting. Radiographic inspections shall be accomplished in accordance with MIL-C-6021. The four mounting flanges shall be Class 1B, Grade B, and the remaining portion of the housing shall be Class 1B, Grade C. The X-ray films and related records shall be arranged to maintain individual casting identities.

4.4.6 Environmental testing (see 3.10).

4.4.6.1 Thermal shock. The switch shall be placed in an approved temperature chamber stabilized at +160 ± 5°F (see 6.4) and the switch shall remain at this temperature for a minimum period of four hours. At the conclusion of this period, and within 5 minutes, the switch shall be transferred to another

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approved chamber stabilized at -60 ± 5°F (see 6.4). The switch shall remain at this temperature for a minimum period of four hours. This constitutes one cycle. The number of complete cycles shall be three. At the conclusion of the third cycle, the switch shall be removed from the test chamber, returned to room temperature, and within a period of one hour test for compliance with 3.10.1.

4.4.6.2 Mechanical shock. The switch shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed on a shock producing device and tested to the levels and tests as specified in 3.10.2.

4.4.6.3 Transportation vibration. The switch shall be mounted securely on a fixture capable of being positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed in a suitable temperature vibration chamber at -65° ± 5°F for 4 hours minimum. The switch shall then be vibrated as specified in 3.11. Upon completion of this test and at extreme temperature, evaluated for compliance with 3.5, 3.6 and 3.7. The switch shall then be prepared and tested as above at a temperature of +160° ± 5°F. Remove the switch from the chamber, allow to return to room temperature, visually examine for external damage (see 6.5) and test for compliance with 3.10.3.

4.4.6.4 Operational vibration. The switch shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted switch shall then be installed on a vibrator and vibrated to the levels and tests as specified in 3.10.4. During vibration testing, contact chatter shall be monitored utilizing approved equipment which will record contact chatter as specified in 3.10.4. After vibration testing, test for compliance with 3.10.4.

4.4.7 Current carrying capacity. The current carrying capacity shall be measured with an approved electrical tester using a voltage of 37 vdc maximum. The maximum voltage drop between the applicable terminal pins of J1 and J2 connectors shall not exceed the value specified in 3.11. The current between the terminal pins shall be 5 ± 0.25 amperes for a duration of fifteen ± one minutes. The unit shall be subjected to a resistive load of 20 ± 2 amperes for 200 ± 5 milliseconds without showing evidence of damage (see 6.5). The unit shall then be tested to determine compliance with 3.11.

4.4.8 Life cycling. With the unit at room temperature mount the open selector on a fixture capable of rotating each switch of the assembly through 300 complete switching cycles. The switch shall meet the requirements of 3.11, 3.5, 3.6 and 3.7 after meeting requirements of 3.12.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The switch shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 External damage. Characteristics such as obliterated markings, damaged finish exposing base metal, damaged wire insulation, rust and cracks that are detectable by visual examination, shall be considered cause for rejection.

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OMB Approval
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MIL-S-63175 (PA)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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